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Patient subtyping in juvenile fibromyalgia: the role of multisensory hypersensitivity and neurophysiological correlates

Laura Martín-Herrero, MSc[1,†], **Maria Suñol, PhD**[1,†], **Saül Pascual-Diaz, PhD**[1], **Tracy V. Ting, MSc**[2], **Jonathan A. Dudley, PhD**[3], **Catherine Jackson**[4], **Susmita Kashikar-Zuck, PhD**[5], **Robert C. Coghill, PhD**[5], **Marina López-Solà, PhD**[6,*]

[1]Institute of Neurosciences, University of Barcelona, Barcelona, Spain; Department of Medicine, School of Medicine and Health Sciences, University of Barcelona, Barcelona, Spain; IDIBAPS, Institut d'Investigacions Biomèdiques August Pi i Sunyer, Barcelona, Spain

[2]Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati, OH, USA; Division of Rheumatology, Cincinnati Children's Hospital Medical Center, Cincinnati, OH, USA

[3]Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati, OH, USA; Division of Behavioral Medicine and Clinical Psychology, Cincinnati Children's Hospital Medical Center, Cincinnati, OH, USA

[4]Division of Behavioral Medicine and Clinical Psychology, Cincinnati Children's Hospital Medical Center, Cincinnati, OH, USA

[5]Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati, OH, USA; Division of Behavioral Medicine and Clinical Psychology, Cincinnati Children's Hospital Medical Center, Cincinnati, OH, USA; Pediatric Pain Research Center, Cincinnati Children's Hospital Medical Center, Cincinnati, OH, United States

[6]Serra Hunter Program, Department of Medicine, School of Medicine and Health Sciences, University of Barcelona, Spain; Institute of Neurosciences, University of Barcelona, Barcelona, Spain; IDIBAPS, Institut D'Investigacions Biomèdiques August Pi i Sunyer, Barcelona, Spain

Abstract

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*Corresponding author: Department of Medicine, School of Medicine and Health Sciences, University of Barcelona, Casanova, 143, Ala Nord, 5a planta, Barcelona 08036, Spain. mlopezsola@ub.edu.

†L.M.-H. and M.S. contributed equally.

Author contributions

Laura Mart'ın Herrero (Conceptualization [lead], Data curation [lead], Formal analysis [lead], Investigation [lead], Writing—original draft [lead]), Maria Suñol (Conceptualization [lead], Formal analysis [equal], Investigation [equal], Supervision [lead], Writing—review & editing [lead]), Saül Pascual-Diaz (Writing—review & editing [equal]), Tracy Ting (Writing—review & editing [equal]), Jonathan A. Dudley (Writing—review & editing [equal]), Catherine Jackson (Data curation [equal], Writing—review & editing [supporting]), Susmita Kashikar-Zuck (Writing—review & editing [equal]), Robert C. Coghill (Writing—review & editing [equal]), and Marina López-Solà (Conceptualization [lead], Funding acquisition [lead], Investigation [equal], Project administration [lead], Supervision [lead], Validation [lead], Visualization [lead], Writing—review & editing [lead])

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Objective: The aim of this study was to investigate whether we could identify groups of adolescents with juvenile fibromyalgia (JFM) based on their subjective perception of non-painful multisensory stimuli in daily life and to study brain function differences between these groups.

Method: 43 female adolescents with JFM (16.56 ± 1.01 years) and 34 healthy controls (16.21 ± 0.89 years) completed validated measures of multisensory hypersensitivity and an fMRI multisensory task. We conducted average linkage cluster analyses, including measures of multisensory sensitivity, and tested between-group differences in core disease-related features, affect-related measures, and task-evoked brain activation in a priori defined regions of interest.

Results: We identified two clusters of participants. The first cluster included 22 patients with higher multisensory sensitivities, and the second one consisted of 21 patients and 34 healthy controls with lower multisensory sensitivities. A second cluster analysis, including only adolescents with JFM, confirmed the same patient division. The two patient subgroups were similar in affective symptoms and coping efficacy but differed in core symptoms of JFM. Adolescents with JFM who experienced higher sensory sensitivities showed increased brain activation in the left primary motor cortex and the left amygdala.

Conclusions: We found two clusters of adolescents with JFM with regard to their non-painful multisensory sensitivities. Greater multisensory hypersensitivity was associated with greater severity of core disease symptoms without compromising affective/cognitive regulation. It was also associated with increased activity of the primary motor cortex and amygdala in response to multisensory stimulation. The study highlights the potential of patient subtyping to understand contributing psychobiological mechanisms.

Keywords

chronic and recurrent pain; adolescents; rheumatology; school-age children

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Chronic pain is highly prevalent among children, affecting approximately one in five children and adolescents (Chambers et al., 2024). Juvenile fibromyalgia (JFM) is a poorly understood chronic widespread pain condition that affects 2%–6% of school-age youth (Kashikar-Zuck et al., 2019). It occurs mostly in female adolescents (Fillingim et al., 2009; Mikkelsson et al., 1997), during a critical developmental period. JFM is characterized by persistent, widespread musculoskeletal pain, fatigue, and sleep and mood disturbances (Häuser et al., 2021; Kashikar-Zuck et al., 2014, 2016; ence symptoms in adulthood, leading to continued physical and psychosocial impairment (Daffin et al., 2020; Kashikar-Zuck et al., 2014, 2016).

The diagnosis of fibromyalgia (FM) has been questioned and historically portrayed as fictitious (Kashikar-Zuck et al., 2014). Even today, in youth, symptoms can be initially viewed as temporary “growing pains” (Kashikar-Zuck et al., 2014). The lack of measurable biological markers and unanimous diagnostic criteria worldwide for JFM complicates the assessment of the disorder, and patients may be referred to numerous pediatric specialists prior to receiving a JFM diagnosis. Despite these limitations, recent research has provided insights into JFM pathophysiology (Kashikar-Zuck et al., 2014; Sil et al., 2014; Suñol et al., 2022, 2024, 2025; Tong et al., 2023), which may have contributed to

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expanding its recognition as a pain syndrome and increasing knowledge regarding its brain pathophysiology.

Given the complex symptom constellation, there is a need for empirically phenotyping/profiling patients based on constructs that may help improve our understanding of the JFM's pathophysiological components, which can then help guide treatment selection on an individualized basis (Edwards et al., 2021). Previous studies have identified subgroups of adult patients with FM based on health-related, psychological, cognitive, and physical characteristics (Ablin et al., 2016; Braun et al., 2020; Fernández-de-Las-Peñas et al., 2023; Sunol et al., 2022). Recent studies provided evidence that non-painful multisensory sensitivities could serve as a relevant feature, as it has already shown the ability to differentiate a range of chronic pain conditions both in adults and adolescents (Walia et al., 2024; Wang & Frey-Law, 2023). Patients with FM and JFM consistently report heightened sensory sensitivity and decreased tolerance to various non-painful sensory stimuli, including auditory, visual, olfactory, taste, and tactile modalities (López-Solà et al., 2014, 2017; Talamilla et al., 2022; Wang & Frey-Law, 2023; Wilbarger & Cook, 2011). A recent study by our group showed that adolescent girls with JFM reported greater daily sensory hypersensitivities and higher unpleasantness in a controlled experimental multisensory stimulation task compared to matched healthy girls (Martín-Herrero et al., 2024). We observed substantial variability—defined as the average distance of each individual to the group mean—among patients in these non-painful multisensory sensitivity measures. Specifically, patients showed 1.71–1.91 times more variability than the control group. This finding suggests that sensory hypersensitivity in JFM may not be uniformly distributed across patients. Given the critical role of multisensory processing in daily function, identifying patient subtypes could enhance our understanding of the multiple mechanisms underlying the condition and potentially inform personalized treatment approaches.

In patients with chronic pain, hyper-unpleasantness and avoidance in response to non-painful sensory events have been associated with increased symptom severity, bodily pain extension, and functional disability (López-Solà et al., 2014; Martín-Herrero et al., 2024). Thus, the presence of such sensory sensitivities may indicate higher impairment in physical, social, and cognitive activities and contribute to dysfunction by potentially increasing stress, sensory overwhelm, anxiety, and fatigue (Wilbarger & Cook, 2011). At the brain level, adults with FM show significant alterations in cortical sensory regions during non-painful multisensory stimulation. Such changes correlate with core FM symptoms (López-Solà et al., 2014, 2017). Whether these alterations are already present in juvenile patients remains to be elucidated. Previous studies have reported altered resting-state connectivity in primary sensory cortices and primary somatosensory-motor (sensorimotor) cortex of patients with juvenile and adult FM (Kim et al., 2015; Pujol et al., 2014; Suñol et al., 2024), suggesting that disrupted cortical sensory and sensorimotor processing may be a central feature of FM. The amygdala plays a critical role in the emotional evaluation of sensory stimuli (Fast & McGann, 2017; Zald, 2003). This region is consistently activated by aversive stimuli across sensory modalities, and its responses are modulated by the arousal and motivational value of the stimuli (Zald, 2003). In adults with FM, the amygdala shows reduced habituation to pain-related fear compared to pain-free individuals and those with other chronic pain conditions (Hsiao et al., 2020). Moreover, this region experiences significant morphological

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and functional changes during adolescence, which may account for heightened emotional reactivity to aversive stimuli (Spear, 2009). This dual vulnerability, stemming from the impact of FM and the developmental stage, makes the study of amygdala responses to non-painful multisensory stimuli in JFM important, particularly because such stimuli are perceived as aversive by patients with JFM.

In this study, we assessed for the first time whether adolescents with JFM can be grouped into different subtypes based on non-painful multisensory sensitivities. We also examined brain response differences between such subtypes during non-painful multisensory stimulation. We hypothesized the emergence of different groups of girls with JFM based on their non-painful multisensory sensitivities. We also hypothesized that girls belonging to the higher multisensory hypersensitivity group may exhibit higher disease-related and affective symptoms and altered multisensory-evoked brain responses in primary sensory and sensorimotor areas (primary visual, primary auditory, and primary sensorimotor cortices) and the amygdala.

Methods

Participants

Seventy-seven adolescent girls participated in the study, including 43 girls diagnosed with JFM and 34 healthy participants. The groups were matched in terms of age (mean ± SD, 16.56 ± 1.01 years for patients vs. 16.21 ± 0.89 years for healthy participants; $t = 1.60$, $p = .11$ [range 14–18 years]), race, caregiver education, household income, and handedness (all right-handed) (p's > .2) (Supplementary Table S1). Data regarding comorbid diseases and medication use are also presented in Supplementary Table S1. Participants were under no medication or under a stable medication regimen for a minimum of 3 weeks prior to the MRI assessment. Pain and psychiatric medications were the most used categories in adolescents with JFM (>40%). We enrolled exclusively female participants because studies based on clinical samples show that JFM is significantly more prevalent in girls (Connelly & Weiss, 2019) and because 86% of eligible JFM participants from the parent clinical trial, the Fibromyalgia Integrative Training program for Teens (FIT-teens) (Kashikar-Zuck et al., 2021), were cisgender girls. Thus, including other genders would have resulted in too small a representation given the possibility of sex/gender differences in pain processing (Bartley & Fillingim, 2013) and brain developmental stage (Lenroot & Giedd, 2006; Menon, 2013). Such differences could have led to relevant, non-addressable confounding effects in the present study.

Inclusion and exclusion criteria are detailed in the document Supplementary Methods S1. Participants provided informed assent, and their parents or legal guardians provided written informed consent. The study protocol and the consent forms were approved by the Ethics and the Institutional Review Board of the Cincinnati Children's Hospital Medical Center (study ID: 2017–7771). Data are available upon reasonable request to the corresponding author. Earlier studies by our group included a partially overlapping patient sample (Suñol et al., 2022, 2024, 2025; Tong et al., 2023). These studies addressed distinct, non-overlapping research questions and employed different fMRI paradigms.

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Measures

To assess core disease symptoms and functional disability, we used developmentally appropriate and validated self-report measures. Participants completed the Pain and Symptom Assessment Tool (PSAT) (Daffin et al., 2020), a standardized tool for the classification of FM in pediatric patients. This tool includes the Fibromyalgia Symptom Severity Scale (FMSS), assessing symptoms experienced in the past week, and the Widespread Pain Index (WPI), reporting body areas where participants experienced pain during the last 3 months (Daffin et al., 2020; Wolfe et al., 2010). Participants self-reported functional disability (i.e., physical difficulties in daily activities) using the Functional Disability Inventory (FDI) (Kashikar-Zuck et al., 2011). Participants also completed the PROMIS Pediatric Fatigue Scale (PROMIS Fatigue) to assesses the level of fatigue in the past week (Varni et al., 2010).

To assess affective symptoms frequently observed in FM, we used developmentally appropriate and validated self-report measures. Participants completed the Child Depression Inventory (CDI) (Kovacs, 1992), a standardized tool for assessing self-reported symptoms of depression in adolescents over the past 2 weeks; the Tampa Scale of Kinesiophobia (TSK) (Tkachuk & Harris, 2012), reporting fear of movement associated with fear of pain; and the Pain Catastrophizing Scale-Children (PCS-C; Crombez et al., 2003) a brief, well-validated 13-item measure assessing pain catastrophizing in children.

To assess potential hypersensitivity to non-painful sensory stimuli in daily life, we used the Adolescent Adult Sensory Profile (AASP) (Brown et al., 2001) and the Questionnaire on Hypersensitivity to Sound (QHS) (Nelting et al., 2002). In the AASP, we focused on two subscales of interest (sensory sensitivity and sensory avoidance) across three sensory modalities of interest (visual, tactile, and auditory). For each modality, a total score was calculated by summing the scores of the two subscales, as is considered standard. These subscales were selected due to their prior successful use in adult patients with FM (López-Solà et al., 2014; Wilbarger & Cook, 2011) and their ability to measure subjective hypersensitivity to daily sensory stimuli. Participants rated the frequency of each item using a five-point Likert scale. Example items included visual (e.g. “I keep the shades down during the day when I am at home”), tactile (e.g., “I’m uncomfortable wearing certain fabrics”), and auditory (e.g., “I startle easily at unexpected or loud sounds”). The QHS measure assesses the subjective distress related to hypersensitivity to sound based on 15 items, under the assumption that the subjective severity of distress is a better indicator of treatment need than audiological findings per se (Nelting et al., 2002).

Multisensory task description

We used a block design fMRI paradigm involving 4 trials of alternating 30-s periods of rest (no stimulation; participants were instructed to relax and focus on a fixation cross) and 30-s periods of multisensory stimulation (López-Solà et al., 2014, 2017) (Supplementary Figure S1). Multisensory stimulation periods consisted of the simultaneous presentation of visual stimulation (at a temporal frequency of 3 Hz [equivalent to six color reversals per second] with a full-field flashing checkerboard composed of a grid of black and white alternating squares [80 ± 10 lux]) and auditory stimulation (a series of 15 tones of frequencies in the

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range of 233.1–1,318.5 Hz, presented at a temporal frequency of 3 Hz, with an intensity of 75 ± 5 dB) coupled with a finger-opposition task during which subjects were instructed to touch the tip of the right thumb with the other fingers (from index finger to little finger). After each multisensory stimulation period and a short interstimulus interval ranging from 2 to 4 s, we asked participants to rate the unpleasantness of the trial (10-s interval), using a computerized visual analogue scale (options ranging from “not at all unpleasant” (0) to “the most unpleasant you can imagine” [100]). At the end of the task, participants also rated bodily pain intensity, unpleasantness, and extension of bodily pain experienced during the entire multisensory task, as well as level of distraction due to bodily pain during the task using similar visual analogue scales.

MRI acquisition parameters

We scanned participants with a Philips Ingenia 3.0-Tesla MRI System (Philips Healthcare, Best, Netherlands) equipped with a 32-channel head coil at the Cincinnati Children’s Hospital Medical Center. The multisensory task BOLD fMRI data were collected using a T2* weighted echo planar imaging sequence with a multiband sensitivity encoding (SENSE) technique and the following scan parameters: multiband acceleration factor = 4, repetition time (TR) = 650 ms, Echo Time (TE) = 30 ms, field of view = 200 mm, flip angle = 53°, voxel size =2.5 × 2.5 × 3.5 mm, slice orientation = transverse (parallel to the orbitofrontal cortex line), slice thickness = 3.5 mm, number of slices = 40 (providing whole-brain coverage), number of volumes = 433, dummy scans = 12, and total scan duration = 4.69 min.

Image processing

Imaging data were processed using gold standard preprocessing pipelines implemented in the CONN Toolbox version 21. a (www.nitrc.org/projects/conn) running on MatLab version 2021a (Math Works Inc, Natick, MA). Standard preprocessing steps included realignment and unwarping, outlier detection, spatial normalization, and smoothing. The first 12 volumes of the multisensory sequence were discarded to allow magnetization to reach equilibrium, and the remaining volumes were realigned with the first one. Based on headmotion parameters estimated during realignment, volumes with framewise displacement >0.9 mm or global BOLD signal changes >5 *SD*s were flagged as outliers. Subsequently, fMRI data were normalized to the 2 mm isotropic MNI space using direct functional normalization, which estimates tissue types and computes the nonlinear transformation directly from the mean BOLD signal image, rather than relying on the anatomical scan (Ashburner & Friston, 2005; Nieto-Castanon, 2020). This approach was chosen both because one participant lacked a T1-weighted image and because prior evidence shows that direct functional normalization reduces inter-subject variability, improves alignment, and increases activation sensitivity (Calhoun et al., 2017; Grabner et al., 2014; Huang et al., 2010). For completeness, we repeated the analyses using an indirect functional normalization method via T1-weighted images (excluding the participant without structural data) (Supplementary Figure S2). This approach was associated with decreased signal integrity, particularly in the prefrontal and temporal cortices. Nevertheless, the results remained consistent across both normalization methods (see Supplementary Table S2). Last, functional images were smoothed with an

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8-mm full-width at half-maximum Gaussian kernel. Images were inspected for artifacts at every step, and quality assurance plots were visually examined.

Statistical analysis

Hierarchical clustering based on sensory hypersensitivities: We performed an average linkage cluster analysis to group the entire sample of 77 adolescents who participated in the study based on their levels of daily sensory hypersensitivity assessed using the AASP (sum of the sensory sensitivity and sensory avoidance scores for each modality, including visual, tactile, and auditory) and the total score of the QHS. We performed an additional average linkage cluster analysis using the same measures but, in this case, including only the adolescents diagnosed with JFM ($n = 43$). Cluster analysis provides a data-driven method to identify meaningful patient subgroups based on inherent patterns in the data (Newcomer et al., 2011). This approach reduces the risk of imposing theoretical assumptions that may not be well-supported in the context of sensory sensitivities in JFM. Average linkage cluster analyses were computed using R software (R Core Team, 2021; R-project.org) to merge groups based on the average distance of all the participants in one group to all the participants in the other. We defined the distance between two clusters as the average distance between all data points in the first cluster and all data points in the second cluster. In both analyses, the silhouette test showed that the number of optimal clusters was two.

fMRI analyses: We ran first-level analyses for each individual participant. We computed general linear models in Statistical Parametric Mapping (SPM) software (SPM12; The Wellcome Department of Imaging Neuroscience) for each participant to identify brain regions with significant activity changes during the multisensory blocks compared to the baseline condition. A primary task regressor was created by convolving the stimulation blocks with a canonical hemodynamic response function. Similarly, we also modeled the unpleasantness question that followed each trial as a regressor. A high-pass filter was used to remove low-frequency signal fluctuations (1/180 Hz). Average framewise displacement and outlier volumes (i.e., volumes with framewise displacement >0.9 mm or global BOLD signal changes >5 *SD*s, which were flagged as outliers during data preprocessing) were regressed out in subject-level analyses.

Contrast images for the task regressor of interest obtained for each subject were carried forward to second-level group analyses. We computed one-sample *t*-tests in SPM12 to obtain group-level activation results. For between-group comparisons, we applied small volume corrections (SVC) on regions of interest (ROIs) determined based on our a priori hypotheses (Supplementary Figure S3). Using the MarsBaR toolbox (marsbar.sourceforge.net), we defined 5 ROIs as 6 mm radial spheres centered at the peak coordinates of the primary visual ($x, y, z_{MNI} = \pm 10, 78, 8$) and auditory ($x, y, z_{MNI} = \pm 58, 14, 8$) cortices, and the left sensorimotor cortex ($x, y, z_{MNI} = -38, -29, 65$), contralateral to the task performed with the right hand, extracted from Sepulcre et al. (2012). The sensory and motor components were combined due to the inherent linkage between sensory processing and motor execution, as well as the involvement of both in the multisensory task. Given the smaller size of the amygdala, we extracted the right and left amygdala ROIs from

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the Brainnetome Atlas (Fan et al., 2016), as it offers a fine-grained functional parcellation of the brain. We found no significant differences at a whole-brain FWE cluster-corrected level of $p < .05$ with voxel-level $p < .001$. Given that this is the first study investigating abnormal brain responses to multisensory stimulation in JFM, we considered an exploratory analysis using a more lenient threshold of $p < .005$ (cluster extent >20 voxels), uncorrected. These results are reported in the online supplementary materials and may provide preliminary insights to guide future studies using larger patient samples. Effect sizes were calculated using Cohen's *D* in R software (R Core Team, 2021; R-project.org).

Results

Two clusters of adolescents with JFM: a cluster of patients whose sensory sensitivity does not differ from healthy adolescents and a cluster of hypersensitive patients

Cluster solution including adolescents with JFM and healthy controls (entire sample): The first cluster analysis included the entire sample ($N = 77$) and revealed two separate clusters. The first one, characterized by higher multisensory hypersensitivities, consisted of 22 patients with JFM (51.16%), and the second one, characterized by lower multisensory hypersensitivity, encompassed 21 patients with JFM (48.84%) and all healthy controls ($n = 34$) (Figure 1A). Significant differences between the two clusters were observed in all variables of interest, with the higher multisensory hypersensitivity cluster showing higher scores in daily sensory sensitivity (AASP visual: $t = 5.16$, AASP touch: $t = 11.34$, AASP auditory: $t = 15.26$, and QHS: $t = 8.90$, all p-values < .0001), core disease symptoms (FDI: $t = 8.13$, WPI: $t = 7.07$, FMSS: $t = 8.41$, and PROMIS Fatigue: $t = 9.86$, all p-values < .0001), and affect-related measures (CDI: $t = 5.10$, TSK: $t = 4.95$, and PCS-C: $t = 4.16$, all p-values < .0001) (Table 1). We identified two JFM participants with mood disorders (anxiety and depression), two with attention-deficit hyperactivity disorder (ADHD), and one with autism spectrum disorder (ASD). We replicated the cluster analysis excluding these five subjects, and the cluster solution remained the same.

Cluster solution including only adolescents with JFM: The second cluster analysis included only participants with JFM and yielded the exact same patient subdivision as the cluster analysis of the entire sample (Figure 1B). Between-group comparisons (adolescents with JFM with higher vs. lower multisensory hypersensitivity) showed significant differences in daily sensory sensitivity (AASP visual: $t = 3.15$, $p = .003$; AASP touch: $t = 6.18$, $p < .0001$; ASSP auditory: $t = 10.67$, $p < .0001$; and QHS: $t = 6.85$, $p < .0001$) and most core disease-related symptoms (FDI: $t = 2.75$, $p = .008$; FMSS: $t = 2.51$, $p = .01$; and PROMIS Fatigue: $t = 5.08$, $p < .0001$). Contrarily, the patient subgroups did not differ in the widespread of pain (WPI: $t = 1.54$, $p = .13$) nor in affect-related measures (CDI: $t = 1.55$, $p = .13$; TSK: $t = 0.65$, $p = .51$; and PCS-C: $t = 0.71$, $p = .48$) (Table 2). Results remained unchanged after replicating the analysis excluding the five JFM participants with either mood disorders, ADHD or ASD.

Brain activation differences during multisensory stimulation

Cluster solution including adolescents with JFM and healthy controls: Adolescents with JFM with higher multisensory hypersensitivity (vs. the lower multisensory

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hypersensitivity group, including patients and controls) showed increased task-evoked brain activation in the left primary motor cortex and the left amygdala ROIs at a SVC-level of FWE peak-corrected level of $p < .05$ and voxel-level $p < .001$ (Figure 2B and Supplementary Table S2). We found no significant differences for the primary visual and primary auditory ROIs. At the whole brain level, we found no significant between-group differences in task-evoked brain activation at an FWE cluster-corrected level of $p < .05$ and voxel-level $p < .001$ (Figure 2A). At an uncorrected level of $p < .005$, cluster extent >20 voxels, adolescents with JFM showing higher multisensory sensitivities showed increased task-evoked brain activation in the left primary motor cortex, and also in primary visual cortices, the right insula, regions of the default mode network (DMN)—bilateral prefrontal cortices and right posterior cingulate cortex—and the amygdala (Supplementary Figure S4A and Supplementary Table S3).

Cluster solution including only adolescents with JFM: Adolescents with JFM with higher multisensory hypersensitivity (vs. adolescents with JFM with lower multisensory hypersensitivity) showed increased task-evoked activation in the left amygdala (SVC, FWE peak-corrected level of $p < .05$ and voxel-level $p < .001$) (Figure 3B and Supplementary Table S2). At the whole brain level, we found no significant between-group differences. At an uncorrected level ($p < .005$, cluster extent >20 voxels) (Figure 3A), we found that adolescents with JFM with higher multisensory hypersensitivity (vs. adolescents with JFM with lower multisensory hypersensitivities) showed increased task-evoked brain activation in the left primary motor cortex, the left amygdala, the right insula, and anterior regions of the DMN (Supplementary Figure S4B and Supplementary Table S4).

Discussion

To our knowledge, this study provides the first evidence of JFM patient subtyping based on non-painful sensory sensitivities. We found that adolescent females with JFM were divided into two groups; one subgroup included approximately 49% of patients who clustered with healthy participants, and they reported lower levels of sensory hypersensitivity, whereas the other 51% of patients were grouped in a separate cluster characterized by higher multisensory hypersensitivities. The two clusters of patients with JFM significantly differed in the severity of core JFM symptoms (present in most patients), including pain, symptom severity (from the PSAT), fatigue, and functional disability; however, the two clusters of patients with JFM did not differ in affective symptoms, including depression, kinesiophobia, and catastrophizing, suggesting that multisensory hypersensitivities in patients with JFM are more strongly associated with sensory and disease-related core symptoms than with affective symptoms. Non-painful multisensory sensitivity has often been associated with FM and might be driven by alterations in central sensory processing (Bar-Shalita et al., 2019), whereas affective symptoms might be influenced by emotional regulation, cognitive biases, or other psychological factors that may not be directly linked to sensory processing. At the brain level, patients belonging to the higher multisensory sensitivity cluster showed increased brain responses to multisensory stimulation in the left primary motor cortex and the left amygdala. At an uncorrected level, such activation increases extended to other sensory processing regions, such as the primary visual cortex and the right insula, and

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to regions of the DMN, which are involved in self-referential processing and attentional shifting.

We found that hyper-unpleasantness and avoidance in response to sensory events in an individual's daily life might contribute to patient stratification. Similarly, previous studies in chronic pain (including FM) have highlighted the significant role of non-painful multisensory sensitivities as a potential transdiagnostic feature that can improve the profiling of pain phenotypes already from a young age and help identify specific neurophysiological mechanisms shared across chronic pain conditions (Walia et al., 2024; Wang & Frey-Law, 2023). Future studies should explore whether multisensory sensitivities in daily life, as measured via questionnaires, are related to heightened sensitivity in quantitative sensory testing measures. Given the link that we observed between higher sensory sensitivity and greater core disease symptoms, future research should investigate whether patients with increased sensory sensitivity are at a higher risk of persistent pain in adulthood and whether the affective domain may also become impacted over time.

At the brain level, adolescents with JFM with higher multisensory hypersensitivity showed overactivation of the left primary motor cortex. This lateralization is congruent with the finger opposition task being performed with the right hand, as motor processing is contralateral. The primary motor cortex plays a prominent role in chronic pain; it is activated in response to peripheral noxious input, and it has been repeatedly associated with pain severity and movement dysfunction (see Chang et al., 2018 for a comprehensive review and meta-analysis). At an uncorrected level, we also found hyperactivations in sensory-processing regions, including the primary visual cortex and the right insula. This brain activation pattern in response to sensory stimulation suggests that in highly sensitive patients, even non-painful sensory stimuli may elicit an amplified neural response. Our findings align with previous reports in adults with FM showing altered cortical sensory processing during multisensory stimulation (López-Solà et al., 2014, 2017). Notably, adult FM studies have reported attenuated responses in early sensory processing regions (e.g., primary sensory cortices) and increased responses at later stages of sensory integration in the insula, which were associated with core FM symptoms. Here, we observed hyperactivation of both primary sensory and sensory integration brain responses in patients with JFM and higher sensory hypersensitivity. Future longitudinal studies are needed to explore these developmental trajectories and their relationship to symptom progression and functional outcomes in FM.

Adolescents with JFM and higher multisensory hypersensitivity also showed hyperactivation of the amygdala in response to multisensory stimulation. While the amygdala is not traditionally classified as a sensory processing region, it receives cortical input from multiple primary sensory cortices, and it projects to brain regions involved in emotional and motivational processing (McDonald, 1998). In this context, the amygdala plays a role in the emotional evaluation of sensory stimuli. In addition, the amygdala is consistently activated by aversive stimuli (Fast & McGann, 2017; Zald, 2003). Our findings may be compatible with the idea that adolescents with JFM and higher sensory sensitivity may perceive non-painful sensory stimuli as more aversive. Potentially amplified emotional evaluation of non-painful sensory input could contribute to the heightened unpleasantness,

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sensitivity, and avoidance observed in patients. Studies in adult FM report amygdala hyperactivation in response to pain-related fear (Hsiao et al., 2020), but not to non-painful multisensory stimulation (López-Solà et al., 2014, 2017). This difference may be linked to the significant neurodevelop-mental changes the amygdala undergoes during adolescence (Spear, 2009), which could influence the neural mechanisms underlying sensory sensitivity in JFM. Relatedly, we found that amygdala responses during pressure pain were more significantly associated with pain reports in adolescents than in adults (Tong et al., 2022), also suggesting a more relevant role of the amygdala in coloring the aversiveness of painful experiences during adolescence.

Lastly, at an uncorrected level, adolescents with JFM and higher sensory hypersensitivity also showed increased activation in regions of the DMN in response to multisensory stimulation. The DMN shows activation decreases during externally oriented attention tasks and activity increases when no particular external task is being attended, and brain resources are allocated to internally oriented self-relevant mentation (Andrews-Hannah et al., 2010; Anticevic et al., 2012). Our findings may suggest that adolescents with JFM and higher sensory hypersensitivity may struggle to appropriately suppress DMN activity when processing external sensory input. This difficulty in deactivating the DMN may indicate increased self-referential processing during sensory input in adolescents, which may be associated with potential shift in the meaning of sensory stimuli for the “self” (Koban et al., 2021). Alternatively, it may reflect a broader disruption in the balance between internally and externally oriented attention during multisensory perception. In line with our findings, multiple studies in adult chronic pain have reported functional alterations in the DMN both at rest and during painful and non-painful sensory tasks (Baliki et al., 2008, 2014; López-Solà et al., 2014, 2017).

This study has some limitations. First, we enrolled only cis-females; thus, our findings cannot be generalized to cis-males, trans, or non-binary patients. Future studies should examine between-sex/gender differences. Second, despite the efforts to recruit a diverse patient profile, our sample is mostly composed of the same ethnicity (white) and participants with medium/high socioeconomic status. Third, for the brain imaging analysis, sample size is a limitation of the current study. Brain imaging findings should be replicated in larger samples.

Fourth, while both normalization approaches yielded consistent results, the choice of normalization method may impact the reproducibility and generalizability of activation patterns in regions susceptible to distortion or signal dropout. Fifth, our task design does not isolate the net contribution of each stimulation modality to the observed results because it was designed to provide a naturalistic presentation of multi-modal sensory stimulation. Future studies aimed at isolating brain processing abnormalities within each modality are warranted. Sixth, our study focuses on three modalities of non-painful sensory stimulation (visual, tactile, and auditory) due to the direct assessment of these three modalities in the multisensory task used in our study and the findings from our previous work (López-Solà et al., 2014, 2017). Movement processing, gustatory, and olfactory modalities might also be clinically relevant in JFM. Finally, this is the first study evaluating the potential of sensory hypersensitivities in subtyping adolescents with JFM. Our findings should be replicated

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in larger patient samples and matched healthy controls to determine their robustness and translational utility.

In conclusion, this study shows that adolescents with JFM experiencing higher multisensory hypersensitivity also have more severe core JFM symptoms without compromising affective/cognitive regulation compared to patients with lower sensory hypersensitivity. In addition, they exhibit increased brain sensory and affective responses to multisensory stimulation, congruent with central sensitization, and an enhanced emotional evaluation of non-painful sensory stimuli. This study highlights the role of patient subtyping to further understand the multiplicity of neurophysiological mechanisms contributing to disease symptoms and the heterogeneity in patient phenotypes.

Relevance for patient care

This study identifies two different JFM patient subtypes regarding whether they show or do not show non-painful multisensory hypersensitivities with potential clinical and neurobiological implications. The results emphasize the need for follow-up studies investigating whether the “multisensory hypersensitive” profile shows a specific clinical trajectory, particularly given the constant exposure to multisensory stimulation in daily life. Early identification of patients with high multisensory sensitivity may be helpful to directly address this problem and prevent its impact on patients’ quality of life and long-term disease outcomes. In line with the study by Walia et al. (2024), patients with sensory hypersensitivities might benefit from a referral to an occupational therapist (OT) if multisensory sensitivities interfere with function. An OT can assess whether interventions, potentially including desensitization, may be appropriate in the specific context.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

Ablin JN, Zohar AH, Zaraya-Blum R, & Buskila D (2016). Distinctive personality profiles of fibromyalgia and chronic fatigue syndrome patients. PeerJ, 4, e2421. [PubMed: 27672497]

Andrews-Hanna JR, Reidler JS, Sepulcre J, Poulin R, & Buckner RL (2010). Functional-anatomic fractionation of the brain’s default network. Neuron, 65, 550–562. [PubMed: 20188659]

Anticevic A, Cole MW, Murray JD, Corlett PR, Wang XJ, & Krystal JH (2012). The role of default network deactivation in cognition and disease. Trends in Cognitive Sciences, 16, 584–592. [PubMed: 23142417]

Ashburner J, & Friston KJ (2005). Unified segmentation. NeuroImage, 26, 839–851. [PubMed: 15955494]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Baliki MN, Geha PY, Apkarian AV, & Chialvo DR (2008). Beyond feeling: Chronic pain hurts the brain, disrupting the default-mode network dynamics. The Journal of Neuroscience: The Official Journal of the Society for Neuroscience, 28, 1398–1403. [PubMed: 18256259]

Baliki MN, Mansour AR, Baria AT, & Apkarian AV (2014). Functional reorganization of the default mode network across chronic pain conditions. PloS One, 9, e106133. [PubMed: 25180885]

Bar-Shalita T, Granovsky Y, Parush S, & Weissman-Fogel I (2019). Sensory modulation disorder (SMD) and pain: A new perspective. Frontiers in Integrative Neuroscience, 13, 27. [PubMed: 31379526]

Bartley EJ, & Fillingim RB (2013). Sex differences in pain: A brief review of clinical and experimental findings. British Journal of Anaesthesia, 111, 52–58. [PubMed: 23794645]

Braun A, Evdokimov D, Frank J, Pauli P, Üçeyler N, & Sommer C (2020). Clustering fibromyalgia patients: A combination of psychosocial and somatic factors leads to resilient coping in a subgroup of fibromyalgia patients. Plos One, 15, e0243806. [PubMed: 33370324]

Brown C, Tollefson N, Dunn W, Cromwell R, & Filion D (2001). The adult sensory profile: Measuring patterns of sensory processing. The American Journal of Occupational Therapy: Official Publication of the American Occupational Therapy Association, 55, 75–82. [PubMed: 11216370]

Calhoun VD, Wager TD, Krishnan A, Rosch KS, Seymour KE, Nebel MB, Mostofsky SH, Nyalakanai P, & Kiehl K (2017). The impact of T1 versus EPI spatial normalization templates for fMRI data analyses. Human Brain Mapping, 38, 5331–5342. [PubMed: 28745021]

Chambers CT, Dol J, Tutelman PR, Langley CL, Parker JA, Cormier BT, Macfarlane GJ, Jones GT, Chapman D, Proudfoot N, Grant A, & Marianayagam J (2024). The prevalence of chronic pain in children and adolescents: A systematic review update and meta-analysis. Pain, 165, 2215–2234. [PubMed: 38743558]

Chang WJ, O'Connell NE, Beckenkamp PR, Alhassani G, Liston MB, & Schabrun SM (2018). Altered primary motor cortex structure, organization, and function in chronic pain: A systematic review and meta-analysis. The Journal of Pain, 19, 341–359. [PubMed: 29155209]

Connelly M, & Weiss JE; for the CARRA Registry Investigators. (2019). Pain, functional disability, and their association in juvenile fibromyalgia compared to other pediatric rheumatic diseases. Pediatric Rheumatology, 17, 1–7. [PubMed: 30611297]

Crombez G, Bijttebier P, Eccleston C, Mascagni T, Mertens G, Goubert L, & Verstraeten K (2003). The child version of the pain catastrophizing scale (PCS-C): A preliminary validation. Pain, 104, 639–646. [PubMed: 12927636]

Daffin M, Gibler RC, & Kashikar-Zuck S (2020). Measures of juvenile fibromyalgia. Arthritis Care & Research, 72, 171–182. 10.1002/acr.24197 [PubMed: 33091238]

Edwards RR, Dworkin RH, Turk DC, Angst MS, Dionne R, Freeman R, & Yarnitsky D (2021). Patient phenotyping in clinical trials of chronic pain treatments: IMMPACT recommendations. Pain Reports, 6, e896. 10.1097/PR9.0000000000000896

Fan L, Li H, Zhuo J, Zhang Y, Wang J, Chen L, Yang Z, Chu C, Xie S, Laird AR, Fox PT, Eickhoff SB, Yu C, & Jiang T (2016). The human brainnetome atlas: A new brain atlas based on connectional architecture. Cerebral Cortex, 26, 3508–3526. [PubMed: 27230218]

Fast CD, & McGann JP (2017). Amygdalar gating of early sensory processing through interactions with locus coeruleus. The Journal of Neuroscience: The Official Journal of the Society for Neuroscience, 37, 3085–3101. [PubMed: 28188216]

Fernández-de-Las-Peñas C, Valera-Calero JA, Arendt-Nielsen L, Martín-Guerrero JD, Cigarán-Méndez M, Navarro-Pardo E, & Pellicer-Valero OJ (2023). Clustering analysis identifies two subgroups of women with fibromyalgia with different psychological, cognitive, health-related, and physical features but similar widespread pressure pain sensitivity. Pain Medicine (Malden, Mass.), 24, 881–889. [PubMed: 36571508]

Fillingim RB, King CD, Ribeiro-Dasilva MC, Rahim-Williams B, & Riley JL III, (2009). Sex, gender, and pain: A review of recent clinical and experimental findings. The Journal of Pain, 10, 447–485. [PubMed: 19411059]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Grabner G, Poser BA, Fujimoto K, Polimeni JR, Wald LL, Trattnig S, Toni I, & Barth M (2014). A study-specific fMRI normalization approach that operates directly on high resolution functional EPI data at 7 Tesla. NeuroImage, 100, 710–714. [PubMed: 24973602]

Häuser W, Brähler E, Ablin J, & Wolfe F (2021). Modified 2016 American College of Rheumatology fibromyalgia criteria, the analgesic, anesthetic, and addiction clinical trial translations innovations opportunities and networks—American Pain Society Pain Taxonomy, and the Prevalence of Fibromyalgia. Arthritis Care & Research, 73, 617–625. [PubMed: 32248629]

Hsiao FJ, Chen WT, Ko YC, Liu HY, Wang YF, Chen SP, Lai KL, Lin HY, Coppola G, & Wang SJ (2020). Neuromagnetic amygdala response to pain-related fear as a brain signature of fibromyalgia. Pain and Therapy, 9, 765–781. [PubMed: 33090368]

Huang C-M, Lee S-H, Hsiao I-T, Kuan W-C, Wai Y-Y, Ko H-J, Wan Y-L, Hsu Y-Y, & Liu H-L (2010). Study-specific EPI template improves group analysis in functional MRI of young and older adults. Journal of Neuroscience Methods, 189, 257–266. [PubMed: 20346979]

Kashikar-Zuck S, Flowers SR, Claar RL, Guite JW, Logan DE, Lynch-Jordan AM, Palermo TM, & Wilson AC (2011). Clinical utility and validity of the Functional Disability Inventory among a multicenter sample of youth with chronic pain. Pain, 152, 1600–1607. [PubMed: 21458162]

Kashikar-Zuck S, Cunningham N, Sil S, Bromberg MH, Lynch-Jordan AM, Strotman D, Peugh J, Noll J, Ting TV, Powers SW, Lovell DJ, & Arnold LM (2014). Longterm outcomes of adolescents with juvenile-onset fibromyalgia in early adulthood. Pediatrics, 133, e592–e600. 10.1542/peds.2013-2220 [PubMed: 24567017]

Kashikar-Zuck S, King C, Ting TV, & Arnold LM (2016). Juvenile fibromyalgia: Different from the adult chronic pain syndrome? Current Rheumatology Reports, 18, 19. [PubMed: 26984803]

Kashikar-Zuck S, Cunningham N, Peugh J, Black WR, Nelson S, Lynch-Jordan AM, Pfeiffer M, Tran ST, Ting TV, Arnold LM, Carle A, Noll J, Powers SW, & Lovell DJ (2019). Long-term outcomes of adolescents with juvenile-onset fibromyalgia into adulthood and impact of depressive symptoms on functioning over time. Pain, 160, 433–441. [PubMed: 30335681]

Kashikar-Zuck S, Briggs MS, Bout-Tabaku S, Connelly M, Daffin M, Guite J, Ittenbach R, Logan DE, Lynch-Jordan AM, Myer GD, Ounpuu S, Peugh J, Schikler K, Sugimoto D, Stinson JN, Ting TV, Thomas S, Williams SE, & Zempsky W; Childhood Arthritis and Rheumatology Research Alliance (CARRA) Pain Workgroup Investigators. (2021). Randomized clinical trial of Fibromyalgia Integrative Training (FIT teens) for adolescents with juvenile fibromyalgia—study design and protocol. Contemporary Clinical Trials, 103, 106321. [PubMed: 33618033]

Kim H, Kim J, Loggia ML, Cahalan C, Garcia RG, Vangel MG, Wasan AD, Edwards RR, & Napadow V (2015). Fibromyalgia is characterized by altered frontal and cerebellar structural covariance brain networks. NeuroImage. Clinical, 7, 667–677. [PubMed: 25844321]

Koban L, Gianaros PJ, Kober H, & Wager TD (2021). The self in context: Brain systems linking mental and physical health. Nature Reviews. Neuroscience, 22, 309–322. [PubMed: 33790441]

Kovacs M (1992). Children's depression inventory. Toronto, Ontario. 10.1037/t19482-000

Lenroot RK, & Giedd JN (2006). Brain development in children and adolescents: Insights from anatomical magnetic resonance imaging. Neuroscience and Biobehavioral Reviews, 30, 718–729. [PubMed: 16887188]

López-Solà M, Pujol J, Wager TD, Garcia-Fontanals A, Blanco-Hinojo L, Garcia-Blanco S, Poca-Dias V, Harrison BJ, Contreras-Rodríguez O, Monfort J, Garcia-Fructuoso F, & Deus J (2014). Altered functional magnetic resonance imaging responses to non-painful sensory stimulation in fibromyalgia patients. Arthritis & Rheumatology, 66, 3200–3209. [PubMed: 25220783]

López-Solà M, Woo CW, Pujol J, Deus J, Harrison BJ, Monfort J, & Wager TD (2017). Towards a neurophysiological signature for fibromyalgia. Pain, 158, 34–47. [PubMed: 27583567]

McDonald AJ (1998). Cortical pathways to the mammalian amygdala. Progress in Neurobiology, 55, 257–332. [PubMed: 9643556]

Martín-Herrero L, Suñol M, Blanc L, Ting TV, Dudley JA, Pascual-Diaz S, Kashikar-Zuck S, Coghill RC, & López-Solà M (2024, June 23–27). Sensory hypersensitivity and symptoms correlates in juvenile fibromyalgia: A multisensory fMRI study. Poster presented at the Organization for Human Brain Mapping Congress, Seoul, Korea. Abstract Book 3: OHBM 2024 Annual Meeting. Aperture Neuro, 495–496. 10.52294/001c.120593

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Menon V (2013). Developmental pathways to functional brain networks: Emerging principles. Trends in Cognitive Sciences, 17, 627–640. [PubMed: 24183779]

Mikkelsson M, Salminen JJ, & Kautiainen H (1997). Non-specific musculoskeletal pain in preadolescents. Prevalence and 1-year persistence. Pain, 73, 29–35. [PubMed: 9414054]

Nelting M, Rienhoff NK, Hesse G, & Lamparter U (2002). Die Erfassung des subjektiven Leidens unter Hyperakusis mit einem Selbstbeurteilungsbogen zur Ger.uschüberempfindlichkeit (GÜF). Laryngo-Rhino-Otol, 81, 327–334.

Newcomer SR, Steiner JF, & Bayliss EA (2011). Identifying subgroups of complex patients with cluster analysis. The American Journal of Managed Care, 17, e324–e332. [PubMed: 21851140]

Nieto-Castanon A (2020). Handbook of functional connectivity magnetic resonance imaging methods in CONN. Hilbert Press.

Pujol J, Macià D, Garcia-Fontanals A, Blanco-Hinojo L, López-Solà M, Garcia-Blanco S, Poca-Dias V, Harrison BJ, Contreras-Rodríguez O, Monfort J, Garcia-Fructuoso F, & Deus J (2014). The contribution of sensory system functional connectivity reduction to clinical pain in fibromyalgia. Pain, 155, 1492–1503. [PubMed: 24792477]

Sepulcre J, Sabuncu MR, Yeo TB, Liu H, & Johnson KA (2012). Stepwise connectivity of the modal cortex reveals the multi-modal organization of the human brain. The Journal of Neuroscience: The Official Journal of the Society for Neuroscience, 32, 10649–10661. [PubMed: 22855814]

Sil S, Arnold LM, Lynch-Jordan A, Ting TV, Peugh J, Cunningham N, Powers SW, Lovell DJ, Hashkes PJ, Passo M, Schikler KN, & Kashikar-Zuck S (2014). Identifying treatment responders and predictors of improvement after cognitive-behavioral therapy for juvenile fibromyalgia. Pain, 155, 1206–1212. [PubMed: 24650858]

Spear LP (2009). Heightened stress responsivity and emotional reactivity during pubertal maturation: Implications for psychopathology. Development and Psychopathology, 21, 87–97. [PubMed: 19144224]

Suñol M, Payne MF, Tong H, Maloney TC, Ting TV, Kashikar-Zuck S, Coghill RC, & López-Solà M (2022). Brain structural changes during juvenile fibromyalgia: Relationships with pain, fatigue, and functional disability. Arthritis & Rheumatology (Hoboken, N.J.), 74, 1284–1294. [PubMed: 35076177]

Suñol M, Dudley J, Payne MF, Tong H, Ting TV, Kashikar-Zuck S, Coghill RC, & López-Solà M (2024). Reduced Cortico-Cortical Resting-State Connectivity in Sensory Systems Related to Bodily Pain in Juvenile Fibromyalgia. Arthritis & Rheumatology (Hoboken, N.J.), 76, 293–303. [PubMed: 37661912]

Suñol M, Pascual-Diaz S, Dudley J, Payne MF, Jackson C, Tong H, Ting TV, Kashikar-Zuck S, Coghill RC, & López-Solà M (2025). Neurophysiology of resilience in juvenile fibromyalgia. Pain. Advance online publication. 10.1097/j.pain.0000000000003562

Talamilla L, Torres-Castro R, Núñez-Cortés R, Sunol M, Solis-Navarro L, & López-Solà M (2022). Sensory system perception/response in people with fibromyalgia versus healthy controls: A systematic review and meta-analysis. PROSPERO.

Tkachuk GA, & Harris CA (2012). Psychometric properties of the Tampa Scale for Kinesiophobia-11 (TSK-11). The Journal of Pain, 13, 970–977. [PubMed: 23031396]

Tong H, Maloney TC, Payne MF, King CD, Ting TV, Kashikar-Zuck S, Coghill RC, & López-Solà M (2022). Processing of pain by the developing brain: Evidence of differences between adolescent and adult females. Pain, 163, 1777–1789. [PubMed: 35297790]

Tong H, Maloney TC, Payne MF, Suñol M, Dudley JA, King CD, Ting TV, Kashikar-Zuck S, Coghill RC, & López-Solà M (2023). Augmented pain-evoked primary sensorimotor cortex activation in adolescent girls with juvenile fibromyalgia. Pain, 164, 2316–2326. [PubMed: 37326678]

Varni JW, Stucky BD, Thissen D, Dewitt EM, Irwin DE, Lai J-S, Yeatts K, & Dewalt DA (2010). PROMIS Pediatric Pain Interference Scale: An item response theory analysis of the pediatric pain item bank. The Journal of Pain, 11, 1109–1119. [PubMed: 20627819]

Walia A, Markfort C, & Frey-Law L (2024). Assessment of multisensory sensitivity may assist with the management of children with chronic pain. Pediatrics, 153, e2023061840. [PubMed: 38577735]

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Wang D, & Frey-Law LA (2023). Multisensory sensitivity differentiates between multiple chronic pain conditions and pain-free individuals. Pain, 164, e91–e102. [PubMed: 35588150]

Wilbarger JL, & Cook DB (2011). Multisensory hypersensitivity in women with fibromyalgia: Implications for well being and intervention. Archives of Physical Medicine and Rehabilitation, 92, 653–656. 10.1016/j.apmr.2010.10.029 [PubMed: 21440712]

Wolfe F, Clauw DJ, Fitzcharles M-A, Goldenberg DL, Katz RS, Mease P, Russell AS, Russell IJ, Winfield JB, & Yunus MB (2010). The American College of Rheumatology preliminary diagnostic criteria for fibromyalgia and measurement of symptom severity. Arthritis Care & Research, 62, 600–610. [PubMed: 20461783]

Zald DH (2003). The human amygdala and the emotional evaluation of sensory stimuli. Brain Research. Brain Research Reviews, 41, 88–123. [PubMed: 12505650]

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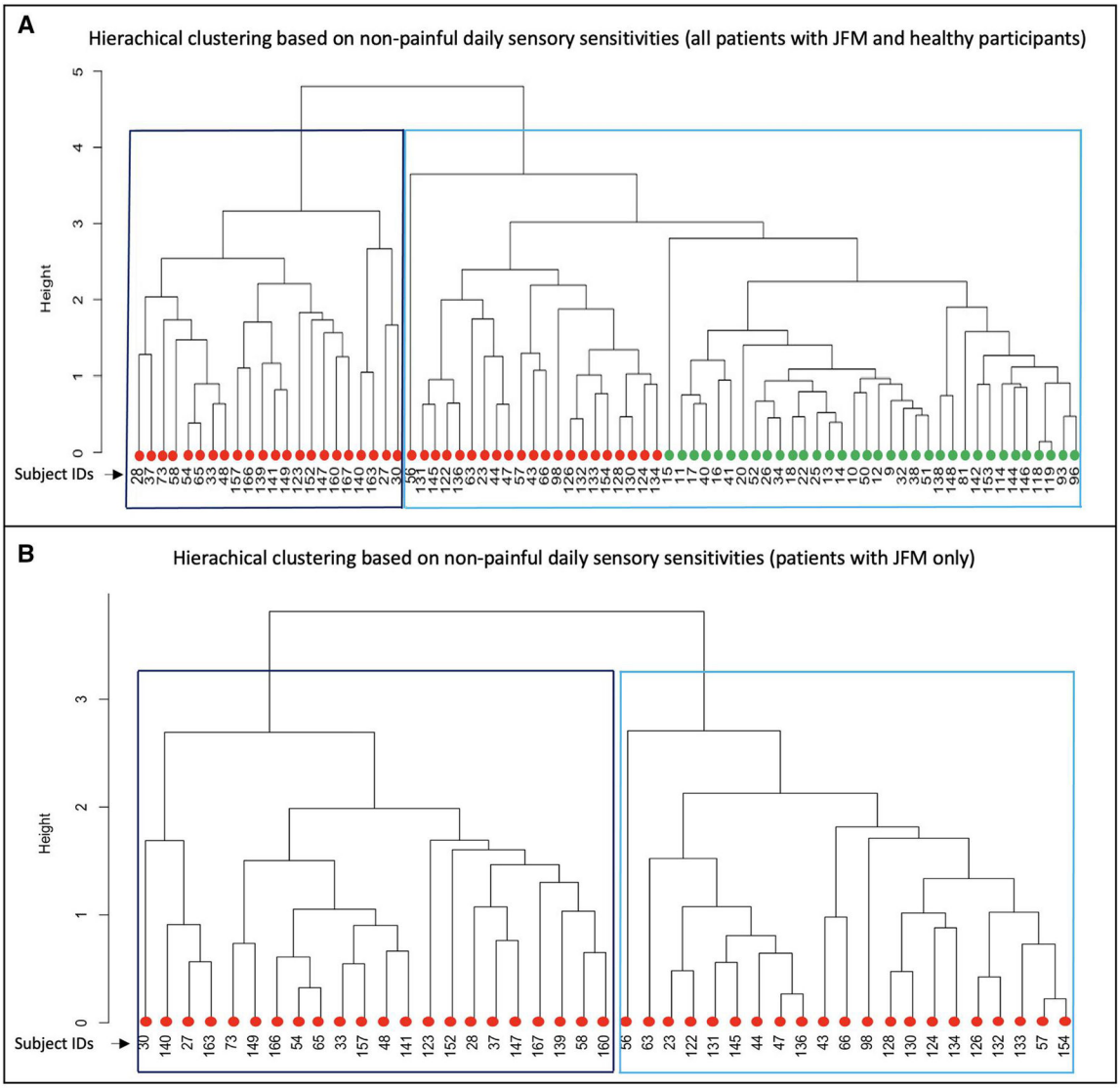

Figure 1.

(A) Hierarchical clustering analysis based on daily non-painful multisensory hypersensitivities including the entire study sample. The cluster on the left comprises participants with higher multisensory hypersensitivities (22 JFM patients) and the one on the right comprises participants with lower multisensory hypersensitivities (21 JFM patients and all healthy controls [n = 34]). (B) Hierarchical clustering analysis, including only patients with JFM. The cluster on the left comprises patients with higher multisensory hypersensitivities (22 JFM patients, the same patients as in [A]) and the one on the right comprises patients with lower multisensory hypersensitivities (21 JFM patients, the same patients as in the [A]). JFM patients are represented with red dots and HC with green dots. The subjects' numbers below were created by the research team to represent each participant of the study, but do not represent any confidential information.

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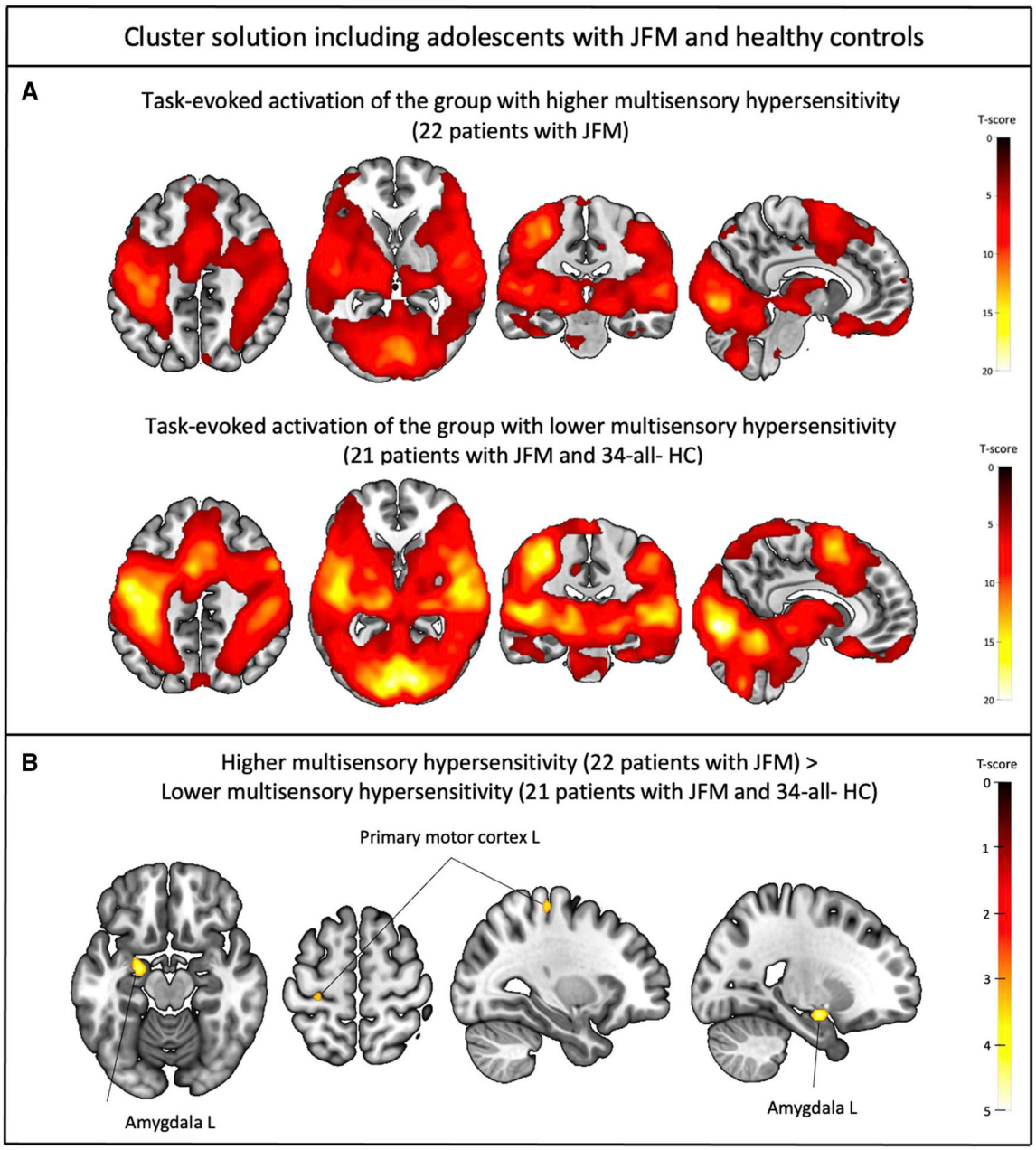

Figure 2.

(A) Task-evoked brain activation at a whole-brain FDR cluster-corrected level of $q < 0.05$ (voxel level, $P < .001$) in participants with higher multisensory hypersensitivities (22 JFM patients) (top) and lower multisensory hypersensitivities (21 JFM patients and 34 HC) (bottom). (B) Differences in brain activation during the multisensory fMRI task between participants with higher multisensory hypersensitivities (22 JFM patients) vs. lower multisensory hypersensitivities (21 JFM patients and 34 HC) applying small volume corrections (SVC) FWE, peak-corrected level of $P < 0.05$ and voxel-level $P < 0.001$ on ROIs determined based on our a priori hypotheses.

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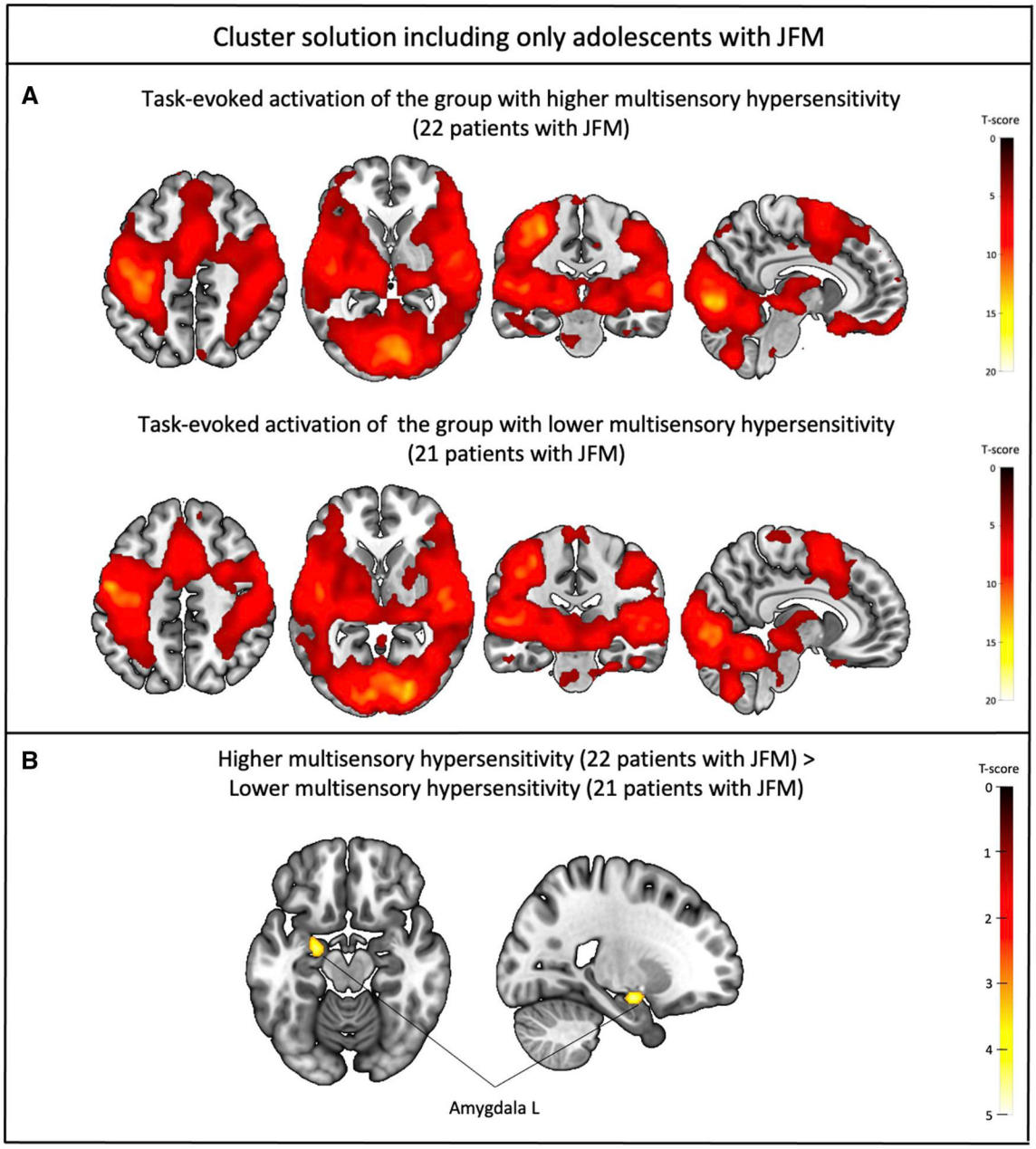

Figure 3.

(A) Task-evoked brain activation at a whole-brain FDR cluster-corrected level of $q < 0.05$ (voxel level $P < 0.001$) in patients with JFM with higher multisensory hypersensitivities ($n = 22$) and patients with lower multisensory hypersensitivities ($n = 21$). (B) Differences in brain activation during the multisensory fMRI task between patients with JFM with higher multisensory hypersensitivities ($n = 22$) and lower multisensory hypersensitivities ($n = 21$) applying SVC FWE peak-corrected level of $P < 0.05$ and voxel-level $P < 0.001$ on ROIs determined based on our a priori hypotheses.

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Table 1.

Differences in sensory sensitivity variables, and core JFM and affective symptoms between participants with higher multisensory hypersensitivity (22 patients with JFM) and participants with lower multisensory hypersensitivity (21 patients with JFM and 34 HC).

	Participants with higher multisensory hypersensitivity (n = 22) Mean (SD)	Participants with lower multisensory hypersensitivity (n = 55) Mean (SD)	Statistics		
			T	p-Value	Degrees of freedom
Sensory sensitivity variables					
AASP visual (0–30)	18.50 (5.49)	11.96 (3.53)	5.16	<.0001	28.24
AASP touch (0–35)	25.63 (3.57)	13.98 (5.10)	11.34	<.0001	54.92
AASP auditory (0–25)	19.68 (2.47)	9.32 (3.15)	15.26	<.0001	49.10
Questionnaire on Hypersensitivity to Sound (0–45)	16.90 (6.94)	3.05 (3.56)	8.90	<.0001	25.54
Core JFM symptoms					
Functional Disability Inventory (0–60)	26.72 (8.43)	8.05 (10.57)	8.13	<.0001	48.28
Widespread Pain Index (0–19)	12.40 (3.20)	4.81 (5.79)	7.07	<.0001	65.87
Fibromyalgia Symptom Severity (0–12)	9.72 (1.51)	4.24 (3.95)	8.41	<.0001	67.94
PROMIS Pediatric Fatigue (0–50)	41.45 (6.89)	21.58 (10.20)	9.86	<.0001	56.89
Affective symptoms					
Child Depression Inventory (0–56)	21.86 (9.09)	9.90 (9.70)	5.10	<.0001	41.14
Tampa Scale of Kinesiophobia (11–44)	27.57 (5.61)	19.73 (7.25)	4.95	<.0001	47.24
Pain Catastrophizing Scale (0–52)	29.61 (11.66)	16.90 (12.14)	4.16	<.0001	38.42

Note. AASP = Adult Adolescent Sensory Profile; HC = healthy control; JFM = juvenile fibromyalgia.

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Table 2.

Differences in sensory sensitivity variables, and core JFM and affective symptoms between patients with JFM with higher multisensory hypersensitivity (n = 22) and lower multisensory hypersensitivity (n = 21).

	JFM with higher multisensory hypersensitivity (n = 22) Mean (SD)	JFM with lower multisensory hypersensitivity (n = 21) Mean (SD)	Statistics		
			T	p-Value	Degrees of freedom
Sensory sensitivity variables					
AASP visual (0–30)	18.50 (5.49)	13.85 (4.07)	3.15	.003	38.704
AASP touch (0–35)	25.63 (3.57)	16.14 (6.10)	6.18	<.0001	32.01
AASP auditory (0–25)	19.68 (2.47)	10.47 (3.12)	10.67	<.0001	38.12
Questionnaire to hypersensitivity to sound (0–45)	16.90 (6.94)	5.28 (3.78)	6.85	<.0001	32.77
Core JFM symptoms					
Functional Disability Inventory (0–60)	26.72 (8.43)	20.14 (7.24)	2.75	.008	40.55
Widespread Pain Index (0–19)	12.40 (3.20)	10.80 (3.58)	1.54	.131	39.97
Fibromyalgia Symptom Severity (0–12)	9.72 (1.51)	8.28 (2.17)	2.51	.01	35.62
PROMIS Pediatric Fatigue (0–50)	41.45 (6.89)	31.42 (6.01)	5.08	<.0001	40.67
Affective symptoms					
Child Depression Inventory (0–56)	21.86 (9.09)	17.14 (10.66)	1.55	.127	39.34
Tampa Scale of Kinesiophobia (11–44)	27.57 (5.61)	26.31 (6.38)	0.65	.51	36.10
Pain Catastrophizing Scale (0–52)	29.61 (11.66)	27.00 (11.28)	0.71	.48	36.42

Note. AASP = Adult Adolescent Sensory Profile; HC = healthy control; JFM = juvenile fibromyalgia.